DE.AUTO

CHAPTER 13

RACING & MEASUREMENT

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

MEMBER EXCLUSIVE · DE.AUTO ·

POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE
MUST-KNOW TERMS

Car modifications, explained

Chapter Thirteen: Racing and Performance Measurement

“NUMBERS DON'T LIE. BUT THEY LOVE CONTEXT.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

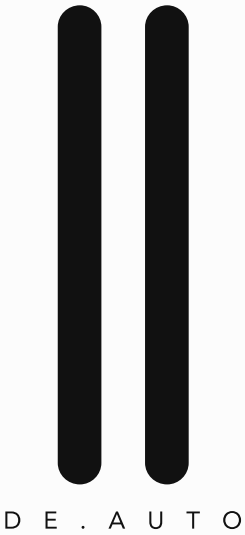
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

DE.AUTO is not responsible for anyone who applies the information in this guide in real life.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

DE.AUTO — Because knowledge is power, and cars are passion.

READY?

A friend says “my car does 0–100 in 4 seconds”… OK, but who timed it? In what weather? On what surface? 🤔

By the end of this chapter you'll know:

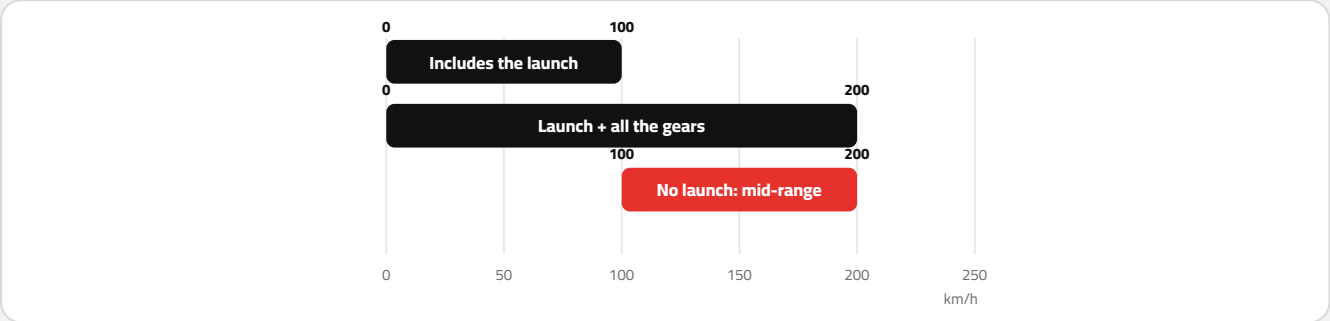
- ✓ The whole language of measuring: 0–100, 100–200, the quarter mile, top speed and braking distance.
- ✓ How the starting tree (Christmas Tree) works, plus the meaning of ET, Trap Speed and 60-Foot.
- ✓ Why does one car launch a full second better, while another flies at the finish?
- ✓ The concept of Launch Control, and why gearbox parts suffer from it.
- ✓ The dyno: wheel horsepower vs engine horsepower, and why the number changes from day to day.
- ✓ Ways to estimate power from time and weight, and how to protect yourself from misleading numbers.
- ✓ The difference between the street and the track, and the basic safety equipment.



01 The language of measuring: every number answers a question

🚦 **Fixed rule:** Everything in this chapter is about **closed tracks and organised events** only. A public road is not a lab or a race track, however empty it looks, and speed or testing there puts you and others at risk and accountable.

A number on its own is like a "cup" without knowing what is inside. That is why each number here has a **question** it answers:



Three measurements on a speed axis: one starts from standstill, and one ignores the launch completely

Measurement	Type	What does it tell you?
0–100 km/h	Acceleration from standstill	The most famous number in advertising; it depends on grip, launch and gearbox more than on raw power
0–200 km/h	Long acceleration	Shows stamina: power, gears and air resistance
100–200 km/h	Acceleration without a launch	Measures mid-range power and gear length; depends on the gear you run it in
¼ mile / ⅛ mile	Standing quarter and eighth mile	Time (ET) + finish speed, from a standing start. The most famous measure at organised short-distance events

402 M is the length of a quarter mile (402.3 m). An eighth of a mile is about half of that: 201.2 m.

✅ **The smart reader's rule:** Before you believe any number, ask 3 questions: **under what conditions** was it measured? **how did the timing start**? And **how many attempts** were made?

📏 **Did you know?** 60 mph = 96.6 km/h. So a 0–60 mph time is slightly shorter than 0–100 km/h for the same car. Watch out when comparing two numbers in different units.

02 Three more measures: top end, stopping, the lap

Top speed

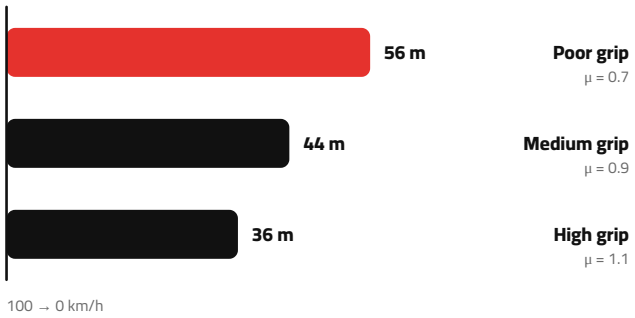
It isn't only about power. At high speeds most of the power **goes into pushing the air aside**, and air resistance rises with the **square** of speed, and the power needed to overcome it rises with its **cube**. So from 200 to 250 km/h, air resistance rises by about **56%**, and you need roughly **double** the power. On top of that, top speed is set by the last gear, the electronic limiter and the tire speed rating.

Braking distance (100–0 km/h)

The question "how much do I need to stop?" matters more than "how fast can I accelerate?". The rule is simple: the distance is proportional to the **square** of speed, and inversely proportional to grip.

$$d = v^2 \div (2 \times \mu \times g)$$

Ideal distance, without the driver's reaction time or brake fade from heat. Illustrative numbers



Theoretical stopping distance from 100 km/h at different levels of grip (μ = approximate grip coefficient)

×4

If you double the speed from 100 to 200 km/h, the theoretical stopping distance becomes **four times as long**, and with it the heat the brakes and tires have to absorb. 🔥 (Brake details are in **Chapter 9**.)

Lap time and the "rolling race" question

Lap time

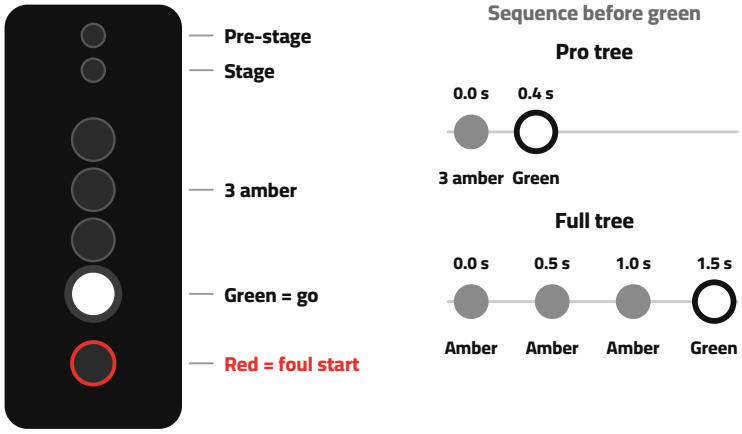
It combines everything: car, driver, tires and track temperature. Compare only on the same track and in the same conditions, never between two different tracks.

Rolling race

A term that, in measuring, means: acceleration starting from a steady speed (such as 100–200), not from standstill. It is not an invitation to race; it is measured on a closed track and only with proper organisation.

03 Short-distance racing: the starting tree

At organised short-distance events, the car starts from standstill and waits for the “tree”: lights that run the start precisely, with sensors that time everything.



The starting tree (a simplified original drawing): in both types, green is the signal to start timing

Reaction time

From the moment of green until the car moves and breaks the sensor. If it moves before that, the red light comes on and it is counted as a “foul start”.

Elapsed time (ET)

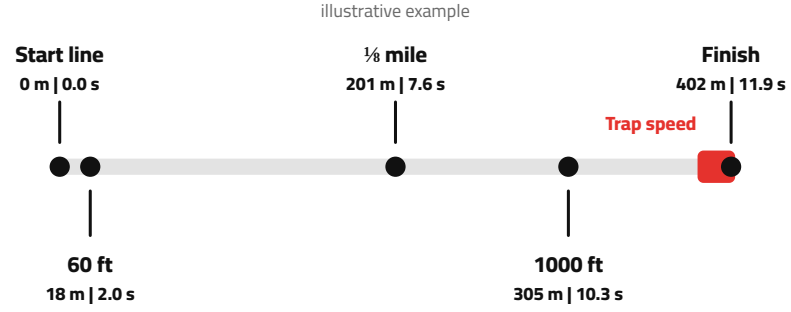
From leaving the start line to the finish line. This is what compares cars (the driver’s reaction time is not included).

Trap speed

The average speed over the last 66 ft (20 m) before the line. The best indicator of overall power.

60-foot time

The time for the first 18 m or so. It shows launch quality (grip). The lower, the better.



The red line is the trap-speed measuring zone

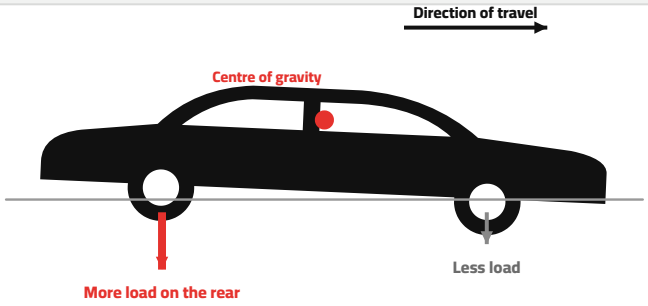
⏱️ **Did you know?** Official timing starts after a small movement (about 30 cm, called Rollout), so the time comes out better than a stopwatch that starts from zero.

04 Quick off the line or quick at the end?

Two cars can cover a quarter mile in the same time: one leaps like a rocket and tires later, the other is slow at first and explodes at the end. The difference is in **who controls the race**.

First seconds: grip decides
- Tires (type, temperature and pressure)
- Weight transfer to the rear
- Drive type (rear / all-wheel)
- Launch revs and how the gearbox copes

Late in the run: power and air decide
- The engine's overall power
- Air resistance and shape
- Gear ratios and their length
- Weight and air density



When you accelerate, weight shifts to the rear (the transfer depends on height, wheelbase and acceleration)

$$\Delta W \approx m \times a \times h \div L$$

The transferred weight grows with mass, acceleration and centre-of-gravity height, and shrinks with a longer wheelbase. Rough example: 1500 kg, 0.8 g, a height of 0.5 m and a wheelbase of 2.7 m → about 220 kg shifts to the rear

How do you read a launch report? 🔍

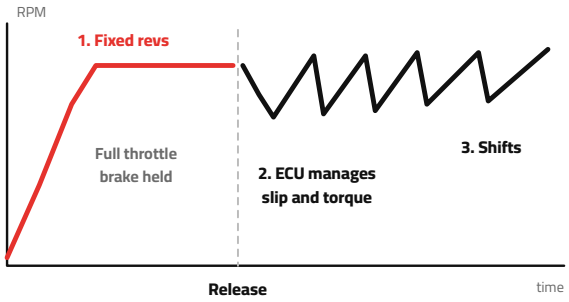
What you see	Likely explanation	Note
Poor 60-ft + high trap speed	A grip problem	The power is there but doesn't reach the ground; tires and setup are the suspects
Excellent 60-ft + low trap speed	Lack of power or short gearing	Left like a rocket, then slowed

🎯 Challenge 1 **A car's 60-foot time is excellent, but its trap speed is low. What is it most likely missing?**

A) Grip | B) Power or suitable gear ratios | C) Driver reaction

05 Launch control: the computer holds the revs

Some modern cars have a mode called **Launch Control** designed for closed tracks. Instead of your foot juggling the throttle and clutch in fractions of a second, **the computer** holds the engine revs at a set number and releases the torque in a controlled way.



Revs against time (illustrative drawing): hold, release, then shifts

How does it work? (the idea only)

1. The computer checks the conditions (gearbox, temperature, pedals) as the manufacturer designed it.
2. It holds the engine revs at the launch figure, with the power ready.
3. On release, it compares wheel speed with car speed and adjusts torque (and spark or boost) to keep a small, useful slip.
4. Once the car settles, the system exits. How it is activated differs from car to car; see the manufacturer's manual and the event rules.

❌ **The hidden cost:** Launching this way throws a sudden load on the clutch (or clutches, in a DCT), driveshafts, differential, engine mounts and tires. That's why some manufacturers set conditions or log usage, and some treat it as outside warranty.

🛑 **Note:** Manufacturer limits exist to protect expensive parts. We don't explain here how to bypass any restriction or counter, and we don't advise trying.

🎯 Challenge 2 **What is the basic idea of a launch control system?**

A) The computer holds the revs and manages slip instead of the driver's foot

B) It always increases engine power

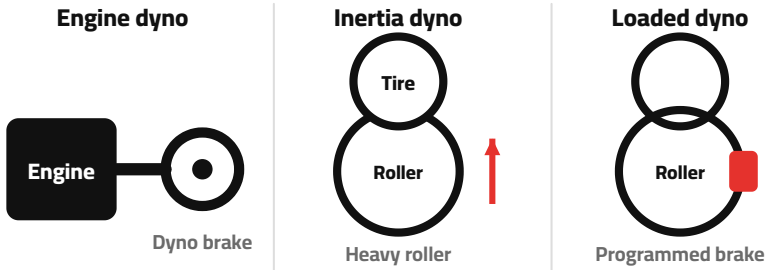
C) It removes the need for tires

06 The dyno: how is power measured?

The dyno (Dynamometer) is a device that measures the **torque** that the engine or the wheels produce, and from torque and revs it calculates power with the formula:

kW = Nm × RPM ÷ 9549

In horsepower: hp = lb-ft × RPM ÷ 5252. Power is calculated from torque, not measured directly



Three common types (simplified drawing)

Engine dyno vs chassis dyno

Engine dyno

The engine sits on a stand connected to a brake. It gives the engine's own figure, is used by manufacturers and builders during development, and is the closest to brochure numbers.

Chassis dyno

The whole car, with its wheels turning on rollers. It measures power at the wheel, including gearbox and differential losses. The most used in workshops.

Inertia dyno vs loaded dyno

Inertia

A heavy roller of known mass; the machine measures how quickly the car accelerates it. Simple and common, but it only measures an accelerating "run".

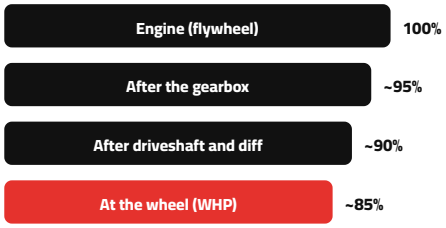
Loaded (brake)

A brake loads the engine so it can hold revs and load. Suits tuning, calibration and testing the engine at a set point.

🔧 **Did you know?** Each type has a different calculation, correction and smoothing method, so comparing numbers between types needs great care.

07 Wheel horsepower vs engine horsepower

You heard “300 hp at the wheels”, and your friend says “350”? Both can be honest, because each one measures from a **different place**.



Example only: about 15% loss; real losses vary from car to car

Power shrinks as it passes through the gearbox, driveshaft, differential and wheels

Engine horsepower

Power at the crankshaft or flywheel. This is what manufacturers usually quote (to a specific measuring standard).

Wheel horsepower

What actually reaches the wheel on a chassis dyno. Usually lower than the engine’s output because of friction and heat in the drivetrain.

How big is the loss? The figures people commonly quote are **10% to 20%**, and it tends to be higher in all-wheel drive and some gearboxes. But these are **general estimates**, not a fixed rule, because loss changes with load, temperature and gearbox oil.

crank ≈ wheel ÷ (1 − loss)

Example: 340 hp ÷ 0.85 ≈ 400 hp (estimate). Note it’s not the same as 340 × 1.15 ≈ 391

✕ MYTH

“Multiply wheel horsepower by 1.15 and you know the engine horsepower exactly.”

✓ FACT

That is only an estimate. The loss changes from car to car and from one measurement to another.

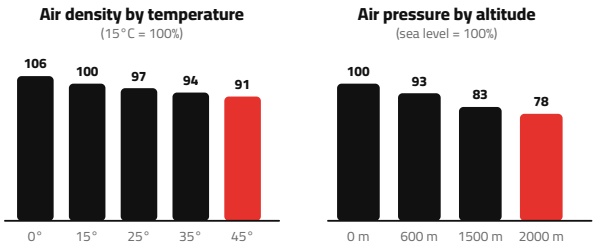
⚖️ **Did you know?** Power units get mixed up: 1 hp ≈ 745.7 W, 1 PS ≈ 735.5 W and 100 kW ≈ 134 hp ≈ 136 PS. A small difference, but it produces two “different” numbers for the same power.

🎯 Challenge 3 **If a chassis dyno gave 270 hp at the wheels, does that mean engine horsepower is below 270?**

A) Yes B) No, usually higher because of the losses C) Exactly the same

08 Why does the same car give different numbers?

The million-dollar question: why does the same car run 11.9 s in the morning and 12.2 in the afternoon? Because **air** is part of the fuel. The engine needs oxygen, and the thinner the air, the less oxygen. Naturally aspirated engines are hit harder than turbo ones (Chapter **7**).



Air density falls with temperature, and pressure falls with altitude (approximate values for a standard atmosphere)

Density altitude

A name that combines temperature, pressure and humidity into one number: "how thin the air is, as if you were at such-and-such altitude". The higher it is, the weaker the engine breathes.

Correction factor

A formula that converts the dyno reading to reference conditions so you can compare. Standards differ on the reference: some use 25°C / 99 kPa and some 20°C / 101.3 kPa.

Other reasons the number changes from dyno to dyno and from day to day

Machine type and calibration, how the car is strapped down, and tire pressure.

The standard used for correction (or none), and the smoothing settings.

Engine and intake-air temperature (Heat Soak): the first pull is not the same as the fifth.

Gear, fan and fuel, battery voltage, and the **operator** themselves.

✅ **Golden rule:** Compare **before and after** on the same machine, in the same mode, and close together in time, and repeat the measurement 3 times. A single number on a piece of paper isn't enough.

🎯 **Challenge 4** **What is the best way to know the effect of a modification on the dyno?**

A) Compare with a number from another machine

B) Same machine, same conditions, before and after, with repeated runs

C) Believe the advertised number

09 Measuring tools and estimating power

Using GPS loggers and phone apps

Instead of a stopwatch, you can measure with a GPS device or an app, but you need to know their limits:

Tool	Strength	Caveat
Ordinary phone	GPS update is usually slow	Fine for comparing yourself with yourself. Lag and how timing starts make a difference
Dedicated GPS logger	Faster updates (usually 10 per second or more)	More accurate for speed and laps, and needs a clear view of the sky
Reading from the car (OBD)	Depends on the car's signal	Can lag or be rounded, and the speedometer usually reads a little higher than the truth
Track timing system	Sensors and gates	This is what is used at organised events

❌ **Don't touch your phone while driving.** Any measurement is done on a closed track with an assistant, with the device mounted securely. If it isn't mounted, don't try.

Estimating power from weight and time

If you have the trap speed or the time, plus the car's mass (with driver and fuel), there are formulas that give an **estimate** of power:

$$kW \approx m \times (V \div 319)^3$$

Trap speed V in km/h, and mass m in kg. Example: 1575 kg and 185 km/h → about 308 kW ≈ 410 hp (estimate)

$$kW \approx m \div (ET \div 6.87)^3$$

Time ET in seconds. Example: 1575 kg and 11.9 s → about 304 kW. This method is affected more by launch quality

$$\text{power-to-weight} = P \div m$$

300 hp on 1500 kg = 200 hp/t, or 5 kg/hp

±10% typical margin of error for these estimates, and it can be larger. The result is "around 410", not exactly 410. Use them for understanding and comparison, not as a promise or a certificate. 🧮

10 Brochure number vs real number

The brochure says “4 seconds” and you got 4.3. Is the car weak? Usually not. The manufacturer measures under certain conditions, and you under others:

Reference conditions
Standard atmosphere, a prepared track, suitable temperature and pressure.

Timing method
Some subtract the Rollout, and some start from the first movement.

Best run
The published figure is usually from a professional driver’s best attempts, not an average.

Units and standard
hp, PS or kW, and net or gross measurement. Fuel, weight and market also make a difference.



Two different numbers doesn’t mean one of them is lying (illustrative numbers)

✕ MYTH
“The number on the dyno is the car’s true power.”

✓ FACT
It is a number from a particular machine on a particular day, in a particular way. Useful for comparing before and after on the same machine, but not a final verdict.

✕ MYTH
“Whoever has the lower 0–100 time is faster overall.”

✓ FACT
0–100 mostly measures the launch more than anything else. A car that beats you in 0–100 may fall behind you in 100–200 or over a lap.

🎯 Challenge 5 **Which one measures “mid-range power” more purely?**

A) 0–100 B) 100–200 C) 60-foot time

11 The track is not the street: and safety rules

	Organised track	The street
The route	One direction, no oncoming cars, pedestrians or signals	Public road: parking, junctions and surprises
Safety zones	Run-off areas and barriers designed for accidents	Street: poles, kerbs and other cars
Load on the car	Brake, oil and tire temperatures are much higher	Daily use is lighter



Track days are organised events with briefings and inspection, and their goal is to learn your car's limits safely. These are the layers of safety

Basic equipment

Approved helmet
To an approved racing safety standard, your size and in date.

Harness + suitable seat
A multi-point harness with a seat designed for it, never fitted by trial and error.

Fire suppression
A mounted extinguisher or a dedicated system fitted by a specialist.

Roll cage
Fitted professionally to a standard, not random welding.

Head and neck protection, plus fire-resistant clothing, gloves and shoes, as the event requires.

Pre-start inspection: tires, bolts, brake fluid and battery mounting, and removal of anything loose.

Flags: each flag has a meaning (stop, caution, faster car behind you, finish). Learn them before you go in.

📄 **Heads-up:** Many insurance policies and dealer warranties exclude track use. Read the terms before any event.

Common mistakes in the world of measuring

❌ **Comparing numbers from different conditions:** a dyno on a different machine, a hot day vs a cold day, or with a different correction.

❌ **Relying on one measurement:** repeat at least 3 times and take the average, and mind the car's temperature between runs.

❌ **Ignoring weight and fuel:** an extra 100 kg (passengers, belongings, petrol) really changes the time.

❌ **Trusting the phone to a hundredth of a second:** apps help for personal comparison; they are not a judge.

❌ **Changing more than one thing at once:** you won't know which one had the effect. Change one thing and measure.

❌ **Testing on the street:** however empty it is. Measuring belongs on a closed track, with organisation and equipment.

Garage talk vs the proper name

Gulf garage talk	Proper name	What it means
Dino / Dinamo	Dynamometer	A device that measures torque and power
Quarter	Quarter Mile	A quarter mile (402 m)
Al-Shajara	Christmas Tree	The launch-start lights
Yahriqha	Red Light / Foul Start	Moved before the green
Reekshin	Reaction Time	From green to first movement
Ay Tee	ET	Total time for the distance
Surat an-nihaya	Trap Speed	Average speed over the last 66 ft
Hisan al-kafar	WHP	Power at the tires
Hisan al-muharrik	BHP / Crank HP	Power at the crankshaft
Launch	Launch Control	Launch control
Reference / Tasheeh	Correction Factor	Converting the number to standard conditions
Data logger	Data Logger	A GPS data logger

Wrap-up 📌

Now you know that a number without context is like a joke without a story. You've seen the language of measuring: 0–100, 100–200, the quarter mile, trap speed and braking distance. You understand the tree, ET, the 60-Foot launch, and the difference between a strong launch and a strong finish. You know the dyno and why numbers change, how to estimate power with caution, and why the track differs from the street. 🚀

🎯 Challenge answers:

1) Power or suitable gear ratios.

2) The computer holds the revs and manages slip instead of the driver's foot.

3) No, usually higher because of the gearbox and differential losses.

4) The same machine and the same conditions, before and after, with repeated runs.

5) 100–200.

➡️ In **Chapter 14: Maintenance and Care**, we close the series with the question that protects everything we have learned: how do you look after your car so it lasts and stays good-looking? Oils, cooling, brakes, tires and a schedule that makes it easy. Ready for the finale? 🔧